

Logical Operators Quick Reference				
$\neg$	$\wedge$	$\vee$	$\rightarrow$	$\leftrightarrow$
$\forall$	$\exists$	$\in$	$\notin$	$\equiv$

- 1A. False – empty sets are a proper subset of every set except for the empty set.
- 1B. True – for any set A, the empty set is a subset of A.
- 1C. False – the only element in the set is $\{\emptyset, \emptyset\}$. Therefore, $\{\emptyset\}$ is not a proper set of the set since the set does not contain $\{\emptyset\}$.
- 1D. False – since $\emptyset$ is not an element of the set, $\{\emptyset\}$ is not a proper set of the set.
- 2A. 0 – the cardinality of an empty set is zero.
- 2B. 1 – the cardinality of $\{\{1, 2, 3\}\}$ is one, despite the inner set containing multiple elements. $\{U\}$ will yield a cardinality of one for a similar reason.
- 2C. 5 – since every element in the set is distinct, the cardinality is the how many outer elements are in the set.
- 2D. 3 – the cardinality of a given set is only the numbers of outer elements if the outer elements are unique. Since $\emptyset$ appears twice in the set, it is only counted once with regards to the cardinality.
- 3A. $\{a, b, c, d, e, f, m, n, o\}$
- 3B. $\{\} = \emptyset$
- 3C. $\{\{m, m\}, \{m, n\}, \{m, o\}, \{n, m\}, \{n, n\}, \{n, o\}, \{o, m\}, \{o, n\}, \{o, o\}\}$
- 3D. $\{\emptyset, \{m\}, \{n\}, \{o\}, \{m, n\}, \{m, o\}, \{n, o\}, \{m, n, o\}\}$
- 3E. $\{\} = \emptyset$
- 3F. 15
- 3G. 2^{256}
- 3H. $\{a, b, c, d, e, m, n, o\}$

4. Lemma: Show that if X , Y , and Z are sets, then $\overline{(X \cap Y)} \cap Z = (\overline{X} \cap Z) \cup (\overline{Y} \cap Z)$.

We will proceed using set builder notation. Let X , Y , and Z be sets and let a be an arbitrary element. We will represent $\overline{(X \cap Y)} \cap Z$ using set-builder notation. The statement can be represented as $\{a \mid a \in Z \wedge a \notin (X \cap Y)\}$, by Definition of Set Builder Notation. We will represent the right hand side of the equation using set builder notation as well. This statement can be represented as: $\{a \mid (a \notin X \cap a \in Z) \cup (a \notin Y \cap a \in Z)\}$, by Definition of Set Builder Notation.

$\{a \mid a \in Z \cap a \notin (X \cap Y)\}$	Given
$\{a \mid (a \notin X \cap a \in Z) \cup (a \notin Y \cap a \in Z)\}$	By Definition of Set Intersection
$\{a \mid (a \in Z \wedge a \notin X) \cup (a \in Z \wedge a \notin Y)\}$	By DeMorgan's Laws
$\{a \mid a \in Z \wedge (a \notin X \vee a \notin Y)\}$	By Distributive Laws
$\{a \mid a \in Z \cap a \notin (X \cap Y)\}$	By Definition of Set Builder Notation
$\{a \mid (a \notin X \cap a \in Z) \cup (a \notin Y \cap a \in Z)\}$	By Distributive Laws
$(\overline{X} \cap Z) \cup (\overline{Y} \cap Z)$	By Simplification

Since we have arrived at the equivalent statement, we conclude our proof.

$$5. P(P(P(\emptyset))) = \{\{\{\emptyset\}, \emptyset\}, \{\{\emptyset\}\}, \{\emptyset\}, \emptyset\}$$

$$P(P(\emptyset)) = \{\{\emptyset\}, \emptyset\}$$

$$P(\emptyset) = \{\emptyset\}$$

$$\text{Therefore, } P(P(P(\emptyset))) - P(P(\emptyset)) = \{\{\{\emptyset\}, \emptyset\}, \{\{\emptyset\}\}, \{\emptyset\}, \emptyset\} - \{\{\emptyset\}, \emptyset\} = \{\{\{\emptyset\}, \emptyset\}, \{\{\emptyset\}\}\}$$

$$\{\{\{\emptyset\}, \emptyset\}, \{\{\emptyset\}\}\} \cup P(\emptyset) = \{\{\{\emptyset\}, \emptyset\}, \{\{\emptyset\}\}, \{\emptyset\}\}$$

$$\{\{\{\emptyset\}, \emptyset\}, \{\{\emptyset\}\}, \{\emptyset\}\} - \emptyset = \{\{\{\emptyset\}, \emptyset\}, \{\{\emptyset\}\}, \{\emptyset\}\}$$

Therefore, we have arrived at our final answer of $\{\{\{\emptyset\}, \emptyset\}, \{\{\emptyset\}\}, \{\emptyset\}\}$.

$$6A. T - (S \cap C)$$

$$6B. (C \cup T) - (C \cap T)$$

$$6C. C - (T \cap S)$$

$$6D. \overline{S \cap T} \cap C$$

$$7. \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}\}$$

8A. We are given that $A \cup (A \cap C) = A$ where A and C are disjoint sets. By definition, A and C share no common elements. Therefore, $A \cap C = \emptyset$. By Substitution, we have $A \cup \emptyset = A$. By Identity Laws, we have $A = A$ and so the set equivalency is proven to be true.

8B. We are given that $P(A) \cup P(B) = P(A \cup B)$ where A , B , and C are disjoint sets. We will prove the proposed set equivalency is false by a counter example.

Let $A = \{1\}$ and $B = \{2\}$. By computation of the lefthand side of the proposed set equivalency, we have $P(A) = \{\emptyset, \{1\}\}$ and $P(B) = \{\emptyset, \{2\}\}$. Therefore, we have $P(A) \cup P(B) = \{\emptyset, \{1\}\} \cup \{\emptyset, \{2\}\}$ which is equivalent to $\{\emptyset, \{1\}, \{2\}\}$.

Let $A = \{1\}$ and $B = \{2\}$, as defined previously. By computation of the righthand side of the proposed set equivalency, we have $A \cup B = \{1, 2\}$. By Substitution, we have $P(\{1, 2\})$ which is equivalent to $\{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$.

Therefore, we have proven the left hand side and the right hand side are not equivalent since $\{\emptyset, \{1\}, \{2\}\} \neq \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$. Therefore, we have proven that the proposed set equivalency is false.

9. We are given $\{9a + 17b \mid a, b \in \mathbb{Z}\} = \mathbb{Z}$ where $\mathbb{Z}$ is the set of integers. We will proceed by showing first that every integer is in the set $\{9a + 17b \mid a, b \in \mathbb{Z}\}$, and every element of $\{9a + 17b \mid a, b \in \mathbb{Z}\}$ is an integer.

We find the GCD of 9 and 17 is 1, which implies there exists integers x and y such that $9x + 17y = 1$. Let n be any integer. Multiplying both sides of the prior equation by n yields $n = n(9x + 17y)$. By Distribution, we have $n = 9xn + 17yn$. If $a = xn$ and $b = yn$, we have $n = 9a + 17b$, thereby showing every integer can be written in the form $9a + 17b$.

Conversely, any element in the form $9a + 17b$ is an integer since a and b are integers and integers are closed under addition and multiplication. Therefore, we have shown that $\{9a + 17b \mid a, b \in \mathbb{Z}\}$ contains every integer and is therefore equal to $\mathbb{Z}$.

10. We will use properties of empty sets to define sets A , B , C , and D as the following:
 $A = \{\} = \emptyset$; $B = \{\} = \emptyset$; $C = \{\} = \emptyset$; $D = \{\} = \emptyset$. The assigning of empty sets is valid since A , B , C , and D simply must be sets; empty sets are sets. We know the cardinality of an empty set is zero. Therefore, we can easily compute $|A| - |B| - |C| - |D| = 0$. Similarly, we know $\emptyset - A = \emptyset$ where A is a set of any type – even an empty set. Therefore, we know $|A - B - C - D| = 0$ since after all calculations, $A - B - C - D = \{\} = \emptyset$.